

PRADIP KRISHNAA MURUGAN

West Lafayette, IN | (765) 631-8750 | pmurugan@purdue.edu | linkedin.com/in/pradipkrishnaa

EDUCATION

Purdue University Jun 2020 – May 2022
Master of Science in Mechanical Engineering **GPA: 4.00**

Purdue University Aug 2016 – May 2020
Bachelor of Science in Mechanical Engineering; Minor in Economics **GPA: 3.61**

EXPERIENCE

Product Engineer Co-op, Cummins Jun 2021 – Present

- Headed validation of MATLAB-based damage simulation software tool to optimize dyno test runtimes for exhaust manifolds
- Performed damage simulations on exhaust manifold heatshield utility for CLAAS 9.0L & Marine 38.0L engines
- Improved physics-based mathematical damage models for power cylinder failure modes with Cummins Technical Area Leader
- Authored a Cummins Engineering Standards (CES) document for Intake and Exhaust Throttle Supplier Test Standards and two Engineering Practice documents on validation test flow and software utility methodology

Research Fellow, Purdue University Jun 2020 – May 2021

- Collaborated with the US DOE to design and develop a modular river current energy turbine
- Chaired the design team through technical development, DFM, DFA, and prototyping phases
- Analysed and implemented topology optimization to reduce system weight by 20% with composites
- Advanced front-end design efficiency with the parameterization of assembly process on Siemens NX
- Improved shaft bending performance with iterative re-design using FEA leading to 3" reduction in bending
- Designed and produced plastic & composite (ABS, Kevlar, Carbon Fiber) parts with FDM 3D Printing, foam airfoil molds with CNC Milling, and metal shafts with CNC Turning using custom CAM setups

Research Assistant, Leibniz University Hannover Apr 2019 – Aug 2019

- Topic: Development of an Interferometric Measuring System for Surface Measurement of Optical Components
- Designed and developed a prototype modular Fizeau Interferometer capable of nanometer resolution
- Reduced total cost of the system by \$2500 and added customizable components to accommodate lenses of varying focal lengths

Product Development Intern, Textron Jan 2019 – Apr 2019

- Directed experiments to improve tank refueling quality by 15% for diesel exhaust fluid tanks
- Implemented a novel heating solution to improve performance of SCR tanks in low temperatures

PROJECTS

Autonomous Lane Following Robot

- Developed an autonomous lane-following robot capable of detecting turns and obstacles
- Analyzed vehicle dynamics to develop PID controllers for the motors to achieve smooth motion
- Designed self-localization systems using IR sensor data processed with LabView

SKILLS

CAD: Siemens NX (Unigraphics), SolidWorks, Fusion 360, Creo, CATIA V5, AutoCAD Inventor

Product Design Tools: NASTRAN (FEM), Design for Assembly/Manufacturing (DFA/DFM), CFD, FEA, GD&T

Simulation & Product Management: MATLAB, LabView, Fusion CAM (Milling, Turning), Siemens Teamcenter

Programming Languages: Python, Julia, C++

Languages: German, English, Tamil, Kannada, French, Arabic

HONORS AND AWARDS

Purdue Graduate Research Assistantship Jun 2020 – May 2021

Gordon and Joanna Hall Memorial Scholarship Aug 2018 – May 2020

Semester Abroad Intercultural Learning Scholarship Jan 2019 – Aug 2019